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An Imaging Genetics Study Based on Brain-wide Genome-wide Association for Identifying Quantitative Trait Loci Related to Pain Sensitivity

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Abstract—**Pain sensitivity has significant individual differences and it is associated with many factors, such as the differentiation of regional structural features of the brain and genetic variation among the population. Until now, a large part of the heritability of pain sensitivity remains unclear. This research focuses on exploring the genetic and neuroimage bases of pain sensitivity. A brain-wide genome-wide association study was carried out on 462 normal subjects, which were divided into high and low pain sensitivity groups according to the cold pain threshold from the cold pressor test. By using voxel-based morphometry (VBM), 116 brain structural features of grey matter (GM) densities were extracted based on high-resolution structural T1-weighted images from magnetic resonance imaging (MRI) scans. Afterward, a genome-wide association study (GWAS) was performed on each phenotype using quality-controlled genotype and analysis data including 755,875 single nucleotide polymorphisms (SNPs). Hierarchical clustering and heat maps were used to demonstrate the GWAS results. Significant associations between SNPs and phenotypes were reported at the threshold ($p < 10^{-6}$). SNPs in the NECTIN1 gene were identified to be strongly associated with various of brain regions, such as the amygdala, hippocampus, and regions at basal ganglia. These results suggest that the imaging genetics study is able to to reveal possible candidate genes and loci that may be associated with pain sensitivity.**

I. Introduction

The advanced brain imaging genetics studies shed light on the research of how genetic variation influences the brain structure and functions [1]. Genome-wide association studies (GWAS) can make use of high-throughput genotyping technology to relate hundreds of thousands of single nucleotide polymorphisms (SNPs) to measurable brain imaging quantitative traits (QTs) [2]. In recent decades, GWAS have rapidly become a powerful tool to reveal risk loci and genes that may be related to human diseases and behavior traits [3].

As one of the most commonly encountered and important symptoms in our everyday lives, pain, however, has been seldom studied using the imaging genetics approach. Pain is a subjective experience and it is highly variable across individuals. Pain sensitivity is believed to be associated with many factors, such as the differentiation of regional structural features of the brain, genetic variation among the population, different psychiatric conditions and personalities. Early research using twin and family studies had established the heritable nature of both experimental and clinical pain, which implies that the variability of pain sensitivity is likely related to some specific genes [4]. Though in recent years, many genes have been proposed to have associations with different types of pain, most studies conducted the analysis based on small groups of candidate genes [5]. On the other hand, since pain perception is processed in the brain, neural mechanism plays an important role in the representation and modulation of pain [6]. To reveal the genetic and neuroimage bases of pain sensitivity, it is of significance to investigate the underlying relationships between genetic variation and brain structure differences, such as gray matter (GM) density from different pain sensitivity groups.

In this study, we present a brain-wide genome-wide association study to explore the brain and genetic architecture of pain sensitivity. GWAS were carried out on 462 normal subjects, which were divided into high and low pain sensitivity groups according to their cold pain threshold from the cold pressor test. The SNP genotyping data were collected using the whole blood of the participants. Accordingly, brain structural images were acquired from a magnetic resonance imaging (MRI) scanner. The GM densities of 116 brain regions were defined and extracted using voxel-based morphometry (VBM). A separate GWAS analysis using PLINK was completed for each of these phenotypes. Afterward, a heat map and hierarchical clustering were used to demonstrate the significant associations between SNPs and phenotypes at threshold $p < 10^{-6}$. Subsequent detailed analysis showed the associations between potential key loci of gene NECTIN1 with several brain regions, such as the amygdala, hippocampus, and basal ganglia. The results suggest that imaging genetic study using brain-wide genome-wide association can reveal novel candidate genes and loci that may be related to pain sensitivity.

II. Materials and Methods

A. Participants

We recruited 462 healthy participants (271 females, age: 21.47 ± 4.23 years) via college and community advertisements for the experiment. The participants were screened before the data acquisition to ensure that they had no history of chronic

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pain, neurological or mental diseases. The participants also had no contraindications to MRI scan.

B. Pain Threshold Measurement

The pain sensitivity of all participants was measured via a cold pressor test with the left hand. Specifically, a participant was asked to firstly immense his/her palm and upper arm into the water with room temperature (22 ± 0.5°C) for 30 seconds. This step was designed to eliminate the skin temperature difference. Then, the participant quickly place the left hand into the cold water (2 ± 0.1°C) and recorded this moment as timepoint 1. When the participant reported that he/she started to feel pain, we recorded this moment as timepoint 2. The duration between these two timepoints were used to measure the cold pain threshold.

Based on the threshold acquired from a cold pressor test, the above participants were firstly sorted from short time (high sensitivity to pain) to long time (low sensitivity to pain), and then equally divided into five groups. To investigate the difference between high sensitivity and low sensitivity of pain, the group in the middle was excluded from this study. As a result, the remaining 370 participants were divided into high and low pain sensitivity groups with equal numbers.

C. Data Acquisition

Structural data were collected using a 3.0T GE MRI scanner. High-resolution structural T1-weighed images were acquired using a three-dimensional magnetization-prepared rapid gradient echo (3D-MPRAGE) sequence. Time of echo is 2.992 ms and time of repetition is 6.896 ms. The size of acquisition matrix is 256×256 and 1×1 mm in-plane resolution. More imaging parameters details can be found in [6].

After MRI data collection, voxel-based morphometry (VBM) [7] was used to process and extract brain-wide target MRI imaging phenotypes. The structural images were registered to the standard Montreal Neurological Institute (MNI) space and segmented into grey matter, white matter, and cerebrospinal fluid. The non-linear deformation parameters were calculated following the algorithm in [8]. The grey matter segments were spatially normalized using the warping function generated from the previous step and then modulated using Jacobian determinant. Finally, the grey matter images were smoothed using the Gaussian kernel with 8 mm full-width-half-maximum. In total, 116 ROI level measurements of mean GM densities were extracted by using the Automated Anatomical Labeling (AAL) atlas [9]. All mean GM density values were referred to as VBM phenotypes.

DNA was isolated from whole blood and extracted using standard procedures. Participants were genotyped with the CapitalBio Technology Corporation Precision Medicine Research Array based on the Affymetrix platform. Genome-wide genotype information was collected at 755,875 SNP markers. To handle possible defects of genetic data, we use the PLINK software (version 1.9) (https://zzz.bwh.harvard.edu/plink/) [10] to perform the quality control of the genotype data. Firstly, only SNPs that locate at autosomes were included in this study and 31,626 SNPs at sex chromosomes were removed. Secondly, SNPs were excluded if they could not meet any of the following criteria: (1) call rate per SNP$\geq$90% (47,481 SNPs removed), (2) minor allele frequency (MAF)$\geq$5% (321,667 SNPs removed), and (3) Hardy-Weinberg equilibrium test of $p \leq 10^{-6}$ (931 SNPs removed). After that, the participants were excluded from this study if the call rate per participant is higher than 90% or the participant cannot pass the gender check. This step had excluded 7 participants from the dataset (6 of them cannot pass the gender check). Population stratification analysis was performed by illustrating the multidimensional scaling (MDS) approach incorporated in PLINK. This method enables us to obtain ethnic information of the sample and to determine possible ethnic outliers (11 participants were excluded). In the end, 352 out of 370 participants and 354,170 out of 755,875 SNP markers remained in our study.

D. GWAS Analysis

The GWAS analysis was conducted individually on 116 imaging phenotypes after the QC procedures of the SNP data. Before that, the phenotypes were adjusted for the baseline age, gender, education level. We apply the PLINK software package to calculate the main effects of all SNPs on the corresponding quantitative imaging phenotype with the quantitative trait association option. An additive SNP effect assumption was used and the empirical p-values were based on the Wald statistics [10].

To investigate the associations between genotype and imaging features of the participants in the following analyses, a heat map was employed to demonstrate the relationship between identified SNPs and the imaging phenotypes at two statistical significance levels: $p < 10^{-6}$ and $p < 10^{-8}$ respectively. A color map was used to represent various significance levels. Traditionally, the association between an SNP and an imaging phenotype can be regarded as significant once it was found that $p < 10^{-6}$. We further included a stricter threshold as $p < 10^{-8}$ to identify the strongest associations. Based on the obtained heat map result, a hierarchical clustering method was applied to identify the similarity among SNPs and imaging phenotypes respectively. We use the Euclidean distance between two nodes to determine the dissimilarity for the hierarchical clustering. The distance average among all pairs of objects in two clusters was used to measure the similarity between two clusters. Two dendrograms were used to show the clustering results for imaging genotypes and SNPs respectively. The association with p-value less than 10^{-6} were recorded for further patter analysis.

Since many previous studies have reported that the amygdala plays a key role in the emotional-affective dimension of pain and for pain modulation, left amygdala mean GM density was selected for detailed sample analysis of a target QT [11]. A Manhattan plot and a quantile-quantile (Q-Q) plot were used to visualize the GWAS results for the left amygdala.

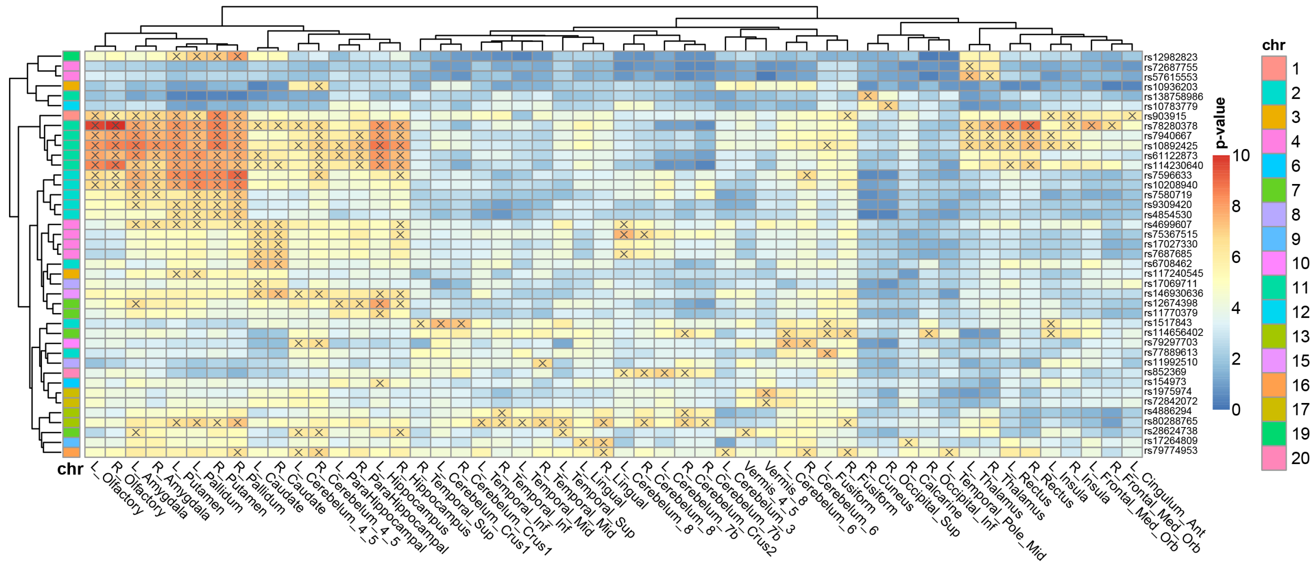

Fig. 1. Heat map of SNP associations with QTs at the $p \leq 10^{-6}$ significance threshold. The blocks with "x" indicate the significance level of $p \leq 10^{-6}$. Only significance SNPs and QTs are shown in the heat map. Dendrograms from x-axis and y-axis derived from heirarchical clusering for QTs and SNPs respectively. The color bars on the left side encode the p-values and corresponding chromosomes for SNPs respectively.

III. Results

A. Sample Characteristics

After QC procedures of the genotyping data, 352 out of 476 recruited participants remained in the following association study. The VBM processing were applied to all these 352 participants to obtained the imaging phenotype data for GWAS analyses. Comparison between the high sensitivity and low sensitivity groups indicated that the gender and education are significantly different (overall $p < 0.05$) between two groups. In the subsequent GWAS analyses, age, gender, education level, handedness, and baseline intracranial volume were included as covariates.

B. GWAS of Imaging Phenotypes

One hundred and sixteen GWAS analyses were applied for all imaging phenotypes. The results are shown as the heat map in Fig. 1 to illustrate the associations among SNPs and the imaging genetics at a significance threshold of $p < 10^{-6}$. The names of identified brain regions and SNPs are shown accordingly. In the figure, a spot with cross symbol "x" indicates such an association have p-value less than 10^{-6}. In total, 237 strong SNP-QT associations were identified at this significance level and 41 SNPs were involved in these associations. Two dendrograms for the hierarchical clustering results of imaging genotypes and SNPs can be found along with the x-axis and y-axis respectively. The color bar on the left side encodes the chromosome IDs for the SNPs.

From the heap map result, we can see that several brain imaging QTs were identified to have statistical significant associations with target SNPs from the GWAS analyses. From the imaging dimension, bilateral amygdala and hippocampus regions can be identified. Also, it is interesting to find that a group of regions from basal ganglia, such as putamen, caudate, olfactory, and pallidum can be identified. It suggests that these regions may be QTs that associated with individual pain sensitivity. Observed from the genomic dimension, several SNPs can be identified from the results, including rs78280378, rs7940667, rs10892425, rs61122873, and rs114230640. All these five SNP are belonging to the same gene named NECTIN1 according to Single Nucleotide Polymorphism Database (dbSNP, https://www.ncbi.nlm.nih.gov/snp/). Among them, rs78280378 and rs10892425 both have the largest number of significant associations with 21 identifications. Additional identified SNPs contain rs10208940 (APLF), rs903915 (MORN1), etc.

The hierarchical clustering results show in Fig. 1 indicate that many pairs of left and right measures of the same region were clustered together as shown in the imaging dimension (x-axis). This suggests the symmetric relationship between these phenotypes and genetic variations. Moreover, several SNP groups from genomic dimension (y-axis) in the same color can be identified besides the aforementioned NECTIN1 gene, including ones from chromosome 2 (rs7596633, rs10208940, rs7580719, rs9309420, rs4854530) and chromosome 4 (rs4699607, rs75367515, rs17027330, rs7687685). This is probably because of the linkage disequilibrium effect that increases the likelihood of non-random association of different alleles.

C. Refined Analysis of Left Amygdala

A target QT, the left amygdala, was selected for a detailed presentation. In Fig. 2, the Manhattan plot of the GWAS for the target QT is shown. The x-axis of the Manhattan plot denotes different chromosomes and the y-axis denotes the p-value of all SNPs. The Manhattan plot indicates that several potential SNPs located at chromosomes 2 and 11 may associate with pain sensitivity. Correspondingly, the Q-Q plot of the left amygdala is shown in Fig. 3. From the figure, we can see that for most of the p-values, the observed p-values from GWAS are almost the same as expected from the null hypothesis. The

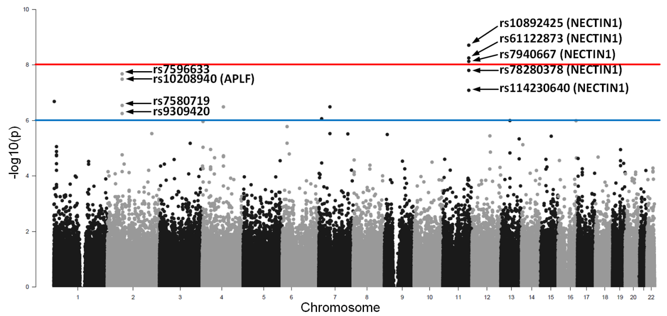

Fig. 2. Manhattan plot of GWAS of QT as the mean GM density of left amygdala. The horizontal lines represent two significance levels: blue line as $p < 10^{-6}$ and red line as $p < 10^{-8}$. Most significant SNPs are marked with their names and corresponding gene if identifiable.

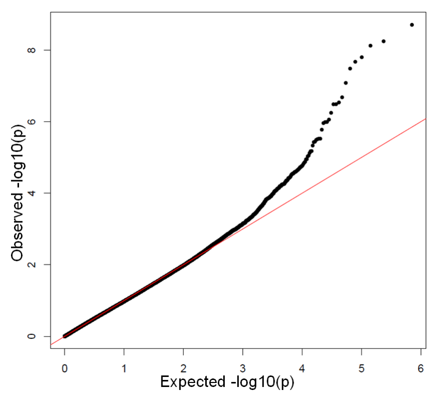

Fig. 3. The Q-Q plot of left amygdala, indicating the distribution of the observed p-values in this sample versus the expected p-values under the null hypothesis of no association.

p-values in the upper tail of the distribution show a significant deviation. It suggests strong associations between these SNPs and the mean GM density of left amygdala.

IV. Discussion

The imaging genetics study conducted in this work has identified many brain regions and SNPs that are probably related to pain sensitivity. For example, strong associations can be found between imaging QTs, such as mean GM density of amygdala, hippocampus as well as regions from basal ganglia and SNPs from NECTIN1 gene. In [12], it was reported that activation in the hippocampus and the amygdala mediates both the anticipation and the experience of an aversive stimulus, such as pain. Both regions are associated with individual differences in response to an upcoming aversive event. Specifically, the hippocampus was reported to be related to pain expectancy sensitivity, whereas the amygdala is associated with the anxiety-related personality trait of harm avoidance [12]. Meanwhile, a lot of research agreed that the basal ganglia regions encode stimulus intensity and play a key role in pain processing and analgesia [13], [14]. The alterations in basal ganglia levels may reflect changes in pain sensitivity [13]. Though it is still far away from a full understanding of individual differences of pain sensitivity, the study showed reasonable results to identify these regions.

On the other hand, from Figs. 1 and 2 we can see that several SNPs from gene NECTIN1 were identified to be associated with the aforementioned brain regions for pain sensitivity. This gene encodes an adhesion protein: calcium(2+)-independent cell-cell adhesion molecule that belongs to the immunoglobulin superfamily. Experiments on rats suggested that NECTIN1 is involved in neuropathic pain. The regulation of NECTIN-1 and its downstream protein c-Src signaling can exert analgesic effects [15]. Our results proposed that the variation of the NECTIN1 gene may influence several brain imaging QTs, i.e. the mean GM density of amygdala, hippocampus, and basal ganglia. It may act as quantitative trait loci that are related to pain sensitivity.

V. Conclusion

This paper introduces an imaging genetics study using brain-wide genome-wide association to investigate possible brain regions and loci that may be related to pain sensitivity. By applying GWAS on 116 mean GM densities as QTs, strong associations of the amygdala, hippocampus as well as basal ganglia regions with several SNPs and genes, such as NECTIN1 can be found. It is possible to deploy imaging genetic research in the future to study the brain imaging and genetic architecture of pain sensitivity.

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